

Ecommerce Analytics Dashboard

Project Overview

The **Ecommerce Analytics Dashboard** is an end-to-end data analytics project designed to analyze sales performance, customer behavior, product performance, payment trends, and business KPIs. The project uses Python for ETL and analysis, and Power BI for interactive dashboarding and reporting.

The dashboard helps stakeholders make **data-driven decisions** by providing insights into revenue growth, customer trends, inventory risks, payment performance, and product sales.

Business Objectives

This project answers the following business questions:

1. What is the total revenue and number of orders?
 2. How many customers are registered?
 3. Which product category performs best?
 4. Which products generate the highest sales?
 5. Which city has the most customers?
 6. Which payment method is used most?
 7. What is the payment success rate?
 8. What is the average order value (AOV)?
 9. How is revenue changing over time?
 10. Which products are at risk of stockout?
-

Dataset Information

The project consists of the following tables:

Customers

Column

CustomerID

Column

CustomerName

Email

MobileNo

City

Country

RegistrationDate

Orders

Column

OrderID

CustomerID

OrderDate

TotalAmount

OrderDetails

Column

OrderDetailID

OrderID

ProductID

Quantity

UnitPrice

Products

Column

ProductID

Column

ProductName

Category

Price

Stock

Rating

CreatedDate

Payments

Column

PaymentID

OrderID

PaymentMethod

PaymentStatus

PaymentDate

KPI Summary

Column

KPIID

ReportDate

TotalRevenue

TotalOrders

TotalCustomers

AvgOrderValue

ETL Process

Extract

- Loaded Excel datasets into Pandas DataFrames.

Transform

- Cleaned missing values.
- Removed duplicate records.
- Converted date fields to datetime format.
- Validated primary and foreign keys.
- Generated business KPIs.
- Created aggregated metrics.

Load

- Loaded transformed data into analytical models and Power BI dashboards.
-

Technologies Used

Tool	Purpose
Python	Data Analysis
Pandas	Data Cleaning & Transformation
NumPy	Numerical Operations
Matplotlib	Data Visualization
Seaborn	Statistical Visualization
Power BI	Dashboard Development
SQL	Data Modeling & Queries
Jupyter Notebook Analysis Environment	

Key KPIs

KPI	Value
Total Revenue	₹13,59,120
Total Orders	70
Total Customers	60
Average Order Value	₹19,416
Payment Success Rate	80%
Peak Revenue Day	18-Jun-2026
Peak Revenue	₹107,320

Key Findings

Revenue Performance

- Generated total revenue of ₹13.59 lakh.
- Revenue showed an upward trend from January to June 2025.
- Highest daily revenue reached ₹107,320.

Product Performance

- Electronics emerged as the best-performing category.
- Samsung Galaxy S24 was the top-selling product.
- Premium electronics contributed significantly to sales.

Customer Analysis

- 60 registered customers.
- Customer base distributed across multiple cities.
- Ahmedabad, Delhi, Noida, Jaipur, Lucknow, Pune, and Mumbai were among the top customer locations.

Payment Analysis

- UPI was the most preferred payment method.
- 80% payment success rate achieved.
- Opportunities exist to reduce failed and pending payments.

Inventory Analysis

- Sony Soundbar
- Brother Printer
- Apple MacBook Air
- Table

These products have the lowest stock levels and require replenishment planning.

Data Model

Relationships used in the project:

- Customers → Orders (1:M)
- Orders → OrderDetails (1:M)
- Products → OrderDetails (1:M)
- Orders → Payments (1:M)

OrderDetails acts as the bridge table between Orders and Products.

Dashboard Pages

Executive Overview

- Revenue KPIs
- Order Metrics
- Customer Metrics
- Payment Summary

Sales Analysis

- Revenue Trends

- Category Performance
- Product Performance

Customer & Product Analysis

- Customer Distribution
- Product Insights

Payment Analytics

- Payment Method Distribution
- Payment Success Analysis

Business Impact

This dashboard provides:

- ☒ Revenue monitoring and growth analysis
- ☒ Product performance tracking
- ☒ Inventory risk identification
- ☒ Customer behavior insights
- ☒ Payment trend monitoring
- ☒ Data-driven business decision support

Future Enhancements

- Real-time dashboard integration
- Customer segmentation analysis
- Sales forecasting using Machine Learning
- Inventory demand prediction
- Customer Lifetime Value (CLV) analysis



Author

Parul Gupta

Project: Ecommerce Analytics Dashboard

Tools: Python | Pandas | NumPy | Matplotlib | Seaborn | Power BI | SQL